Welcome to the Bob wxo MCP servers lab!

You will learn how to:

- Install and configure watsonx Orchestrate ADK to interact with a watsonx Orchestrate instance
- Configure Bob MCP servers to work with the watsonx Orchestrate instance from Bob
- Create an agent using a simple tool project with Bob in watsonx Orchestrate

Prerequisites: Install wxo ADK

Join the Cloud

In your email inbox, you have received an invitation to join the Cloud. Follow the instructions in the email to join the Cloud.

IBM **Cloud**

Account

IBM Cloud Logo

Action required: You are invited to join an account in IBM Cloud

Source ID: 1778517788252
Type: Account
Update Time: 11 May 2026, 4:43 PM UTC

View notification (in subscribed accounts):

Hi nicolas.sapin@fr.ibm.com,

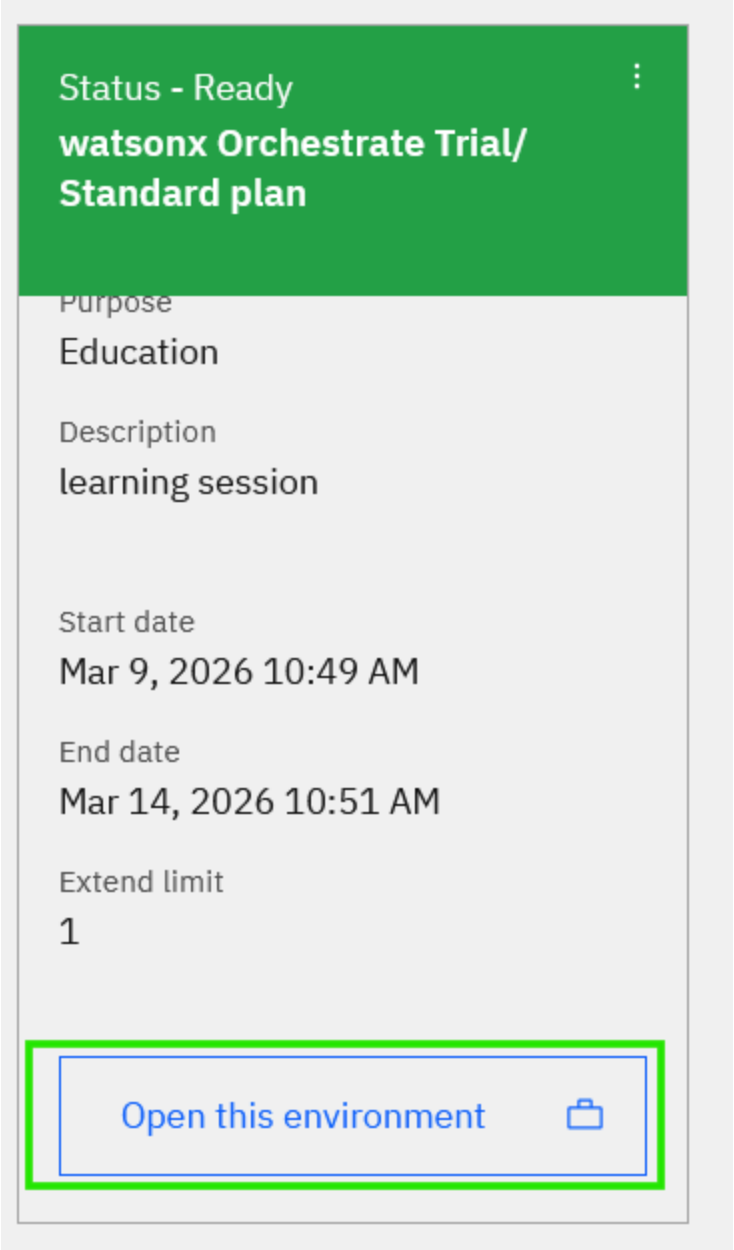
IBM IBM Technology Zone Adhoc 01 invited you to join the following IBM Cloud account: itz-watsonx-031 (Account ID: 5c7165eca27b4d538ce71dda3e3c6799).

Join now.
The link to join the account expires in 30 days.

Thank you,
IBM Cloud

In case of any issues, please contact IBM Cloud support.

Get connected to techzone to check the status of your instance.



IBM Technology Zone | My TechZone | Support

Education

Opportunity Product(s)

Opportunity description
learning session

Customer(s)

Environment

Reservation ID
69ae979f57387b3c23da98cf

Type
IBM Cloud

Request method
itz-events-saas-idp

Transaction ID
ba6c40ad-1603-47de-97b7-e97038062270

Cloud Account
ITZ-WATSONX-029

Geo
americas

Region
us-south

Datacenter
dal10

Customer data
false

Environment
itz-wxo-69ae979f57387b3c23da98

Idle runtime limit
10800

Timeout action

Reservation Details

IBM Cloud Login
https://cloud.ibm.com/resources

Username
<your IBMID>

Password
<your ibm cloud password>

Check the cloud number in the top-right corner of your resource list page.

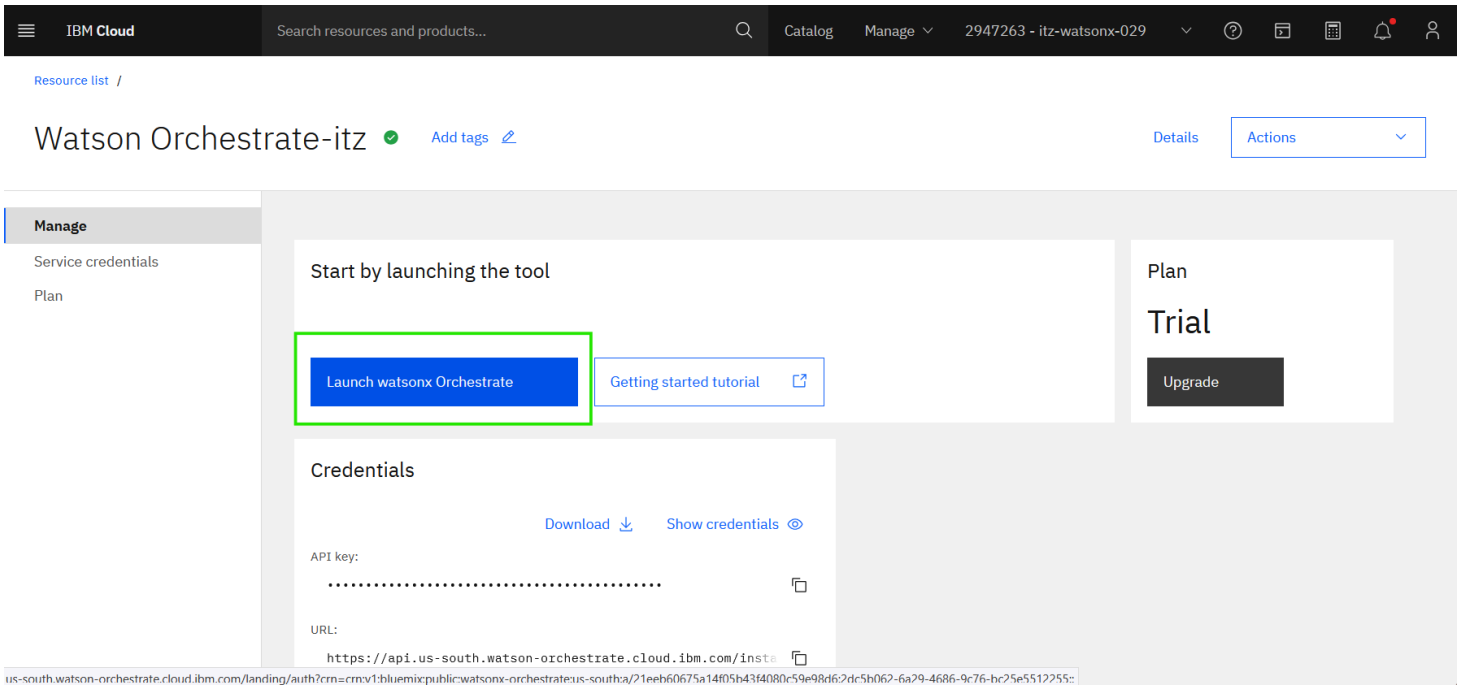
IBM Cloud | Search resources and products... | Catalog | Manage | 2947263 - itz-watsonx-029

Resource list

Create resource +

Name	Group	Location	Product	Status	Tags
Filter by name or IP address...	Filter by group...	Filter...	Filter...	Filter...	Filter...
AI / Machine Learning (4)					
Watson OpenScale-itz	watsonx	Dallas (us-south)	watsonx.governance	active	–
Watson Orchestrate-itz	itz-wxo-69ae979f57387b3c23da98	Dallas (us-south)	watsonx Orchestrate	active	–
Watson Orchestrate-itz 387b3c23da98	itz-wxo-69ae979f57387b3c23da98	Dallas (us-south)	watsonx.ai Runtime	active	–
ws-itz-wxo-69ae979f57387b3c23da98	itz-wxo-69ae979f57387b3c23da98	Dallas (us-south)	watsonx.ai Studio	active	–
Analytics (1+)					
Application development (0)					
Asset management (0)					
Blockchain (0)					
Business automation (0)					
Compute (0)					

cloud.ibm.com/services/watsonx-orchestrate/crn:v1:bluemix:public:watsonx-orchestrate:us-south:a/21eeb60675a14f05b43f4080c59e98d6:2dc5b062-6a29-4686-9c76-bc25e5512255::



Install uv (if not already installed)

Windows:

To install uv on Windows, open PowerShell or Command Prompt and run:

```
powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

Alternatively, you can use pip:

```
pip install uv
```

Mac:

To install uv on Mac, open Terminal and run:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Or using Homebrew:

```
brew install uv
```

Or using pip:

```
pip install uv
```

After installation, verify uv is installed correctly by running:

```
uv --version
```

Install ADK

The current ADK requires Python 3.11 or later; anything older than 3.11 is not supported for installation.

Create a Python virtual environment with this command:

```
python -m venv venv
```

Activate the virtual environment:

Windows (CMD):

```
venv\Scripts\activate.bat
```

Windows (PowerShell):

```
venv\Scripts\Activate.ps1
```

Mac/Linux:

```
source venv/bin/activate
```

After activation, upgrade pip and install the ADK:

```
pip install --upgrade ibm-watsonx-orchestrate
```

```
Windows PowerShell
PS C:\Users\082037706\Documents\2026\bob\lab> python -m venv venv
PS C:\Users\082037706\Documents\2026\bob\lab> venv\Scripts\Activate.ps1
(venv) PS C:\Users\082037706\Documents\2026\bob\lab> pip install --upgrade ibm-watsonx-orchestrate
Collecting ibm-watsonx-orchestrate
  Downloading ibm_watsonx_orchestrate-2.5.1-py3-none-any.whl.metadata (1.4 kB)
Collecting certifi>=2024.8.30 (from ibm-watsonx-orchestrate)
  Downloading certifi-2026.2.25-py3-none-any.whl.metadata (2.5 kB)
Collecting charset-normalizer>=2 (from ibm-watsonx-orchestrate)
  Downloading charset_normalizer-3.4.5-cp312-cp312-win_amd64.whl.metadata (39 kB)
Collecting click<8.2.0,>=8.0.0 (from ibm-watsonx-orchestrate)
  Using cached click-8.1.8-py3-none-any.whl.metadata (2.3 kB)
Collecting docstring-parser<1.0,>=0.16 (from ibm-watsonx-orchestrate)
  Using cached docstring_parser-0.17.0-py3-none-any.whl.metadata (3.5 kB)
Collecting httpx<1.0.0,>=0.28.1 (from ibm-watsonx-orchestrate)
  Using cached httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Collecting ibm-cloud-sdk-core>=3.24.2 (from ibm-watsonx-orchestrate)
  Downloading ibm_cloud_sdk_core-3.24.4-py3-none-any.whl.metadata (8.7 kB)
Collecting jsonref==1.1.0 (from ibm-watsonx-orchestrate)
  Using cached jsonref-1.1.0-py3-none-any.whl.metadata (2.7 kB)
Collecting langchain-core~=1.2.4 (from ibm-watsonx-orchestrate)
  Downloading langchain_core-1.2.17-py3-none-any.whl.metadata (4.4 kB)
Collecting packaging>=24.2 (from ibm-watsonx-orchestrate)
  Using cached packaging-26.0-py3-none-any.whl.metadata (3.3 kB)
Collecting pydantic<3.0.0,>=2.10.3 (from ibm-watsonx-orchestrate)
  Using cached pydantic-2.12.5-py3-none-any.whl.metadata (90 kB)
Collecting pyjwt<3.0.0,>=2.10.1 (from ibm-watsonx-orchestrate)
  Using cached pyjwt-2.11.0-py3-none-any.whl.metadata (4.0 kB)
Collecting python-dotenv>=1.0.0 (from ibm-watsonx-orchestrate)
  Downloading python_dotenv-1.2.2-py3-none-any.whl.metadata (27 kB)
Collecting pytz>=2025.2 (from ibm-watsonx-orchestrate)
```

Configure the ADK Environment

Create the "lab" environment in the ADK with:

```
orchestrate env add -n lab -u https://api.us-south.watson-orchestrate.cloud.ibm.com/instances/2c
```

Activate the ADK environment with:

```
orchestrate env activate lab -a GTLI6TuscI1DBJd0EwQbJbTAzpNCrqAQA1JDKPj0ARSn
```

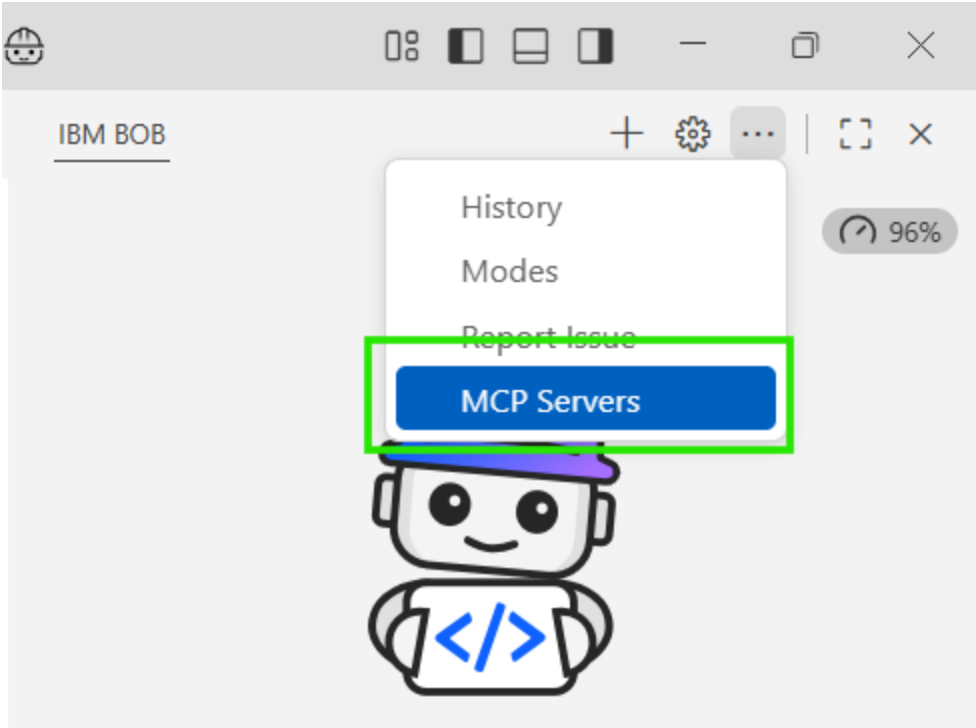
And test it with:

```
orchestrate agents list
```

```
Windows PowerShell
(venv) PS C:\Users\082037706\Documents\2026\bob\lab> orchestrate env add -n lab -u https://api.us-south.watson-orchestrate.cloud.ibm.com/instances/2dc5b062-6a29-4686-9c76-bc25e5512255
[INFO] - Environment 'lab' has been created
(venv) PS C:\Users\082037706\Documents\2026\bob\lab> orchestrate env activate lab -a GTLI6TuscI1DBJd0EwQbJbTAzpNCrqAQA1JDKPj0ARSn
[WARNING] - Using 'ibm_iam' Auth Type. If this is incorrect please use the '--type' flag to explicitly choose one of mcsp, mcsp_v1, mcsp_v2 or cpd
[INFO] - Environment 'lab' is now active
(venv) PS C:\Users\082037706\Documents\2026\bob\lab> orchestrate agents list
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
(venv) PS C:\Users\082037706\Documents\2026\bob\lab>
```

Install and Configure the wxo MCP Servers in Bob

Go to Bob MCP servers:



Install the watsonx Orchestrate ADK Docs MCP server:

Search settings... Ctrl+F

- General
- Chat
- Editor
- Bob findings
- MCP
- Modes

Version: 1.0.0

- Bob Docs
- Report issue

The Terraform MCP Server provides seamless integration with Terraform Registry APIs, HCP Terraform, and Terraform Enterprise, enablin...

Provides time and timezone conversion capabilities using IANA timezone names, with automatic system timezone detection and...

Vault Install

Author: hashicorp

The Vault MCP Server provides integration with HashiCorp Vault for managing secrets and mounts, including KV v1/v2, PKI...

IBM Data Intelligence Install

Author: IBM

IBM Data Intelligence MCP Server provides modular and scalable integration with IBM Data Intelligence services, enabling secure...

WatsonX Flows Install

Author: IBM

Enables integration with watsonx.ai Flows Engine, providing tools for Google Books and Wikipedia searches through a TypeScript-...

watsonx Orchestrate ADK D... Install

Author: IBM

Provides a SearchIbmWatsonxOrchestrateAdk tool which is able to query the watsonx Orchestrate ADK Developer documentation....

watsonx Orchestrate ADK Install

Author: IBM

Give Bob direct access to all commands within the watsonx Orchestrate ADK. Empowers Bob to conversationally author, import or list...

IBM watsonx.data Install

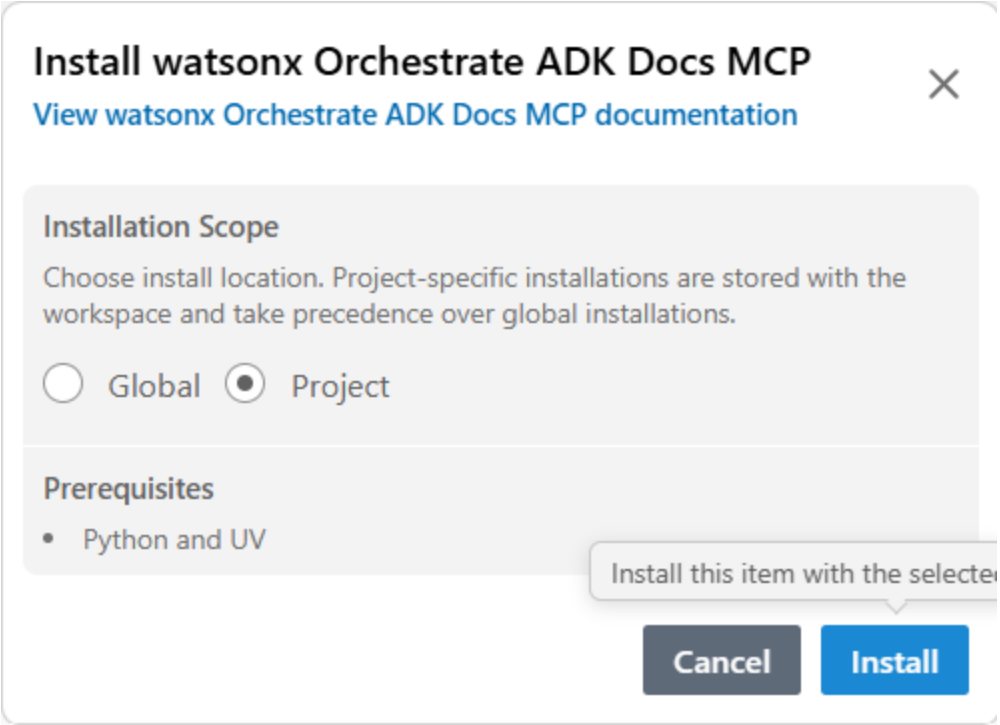
Author: IBM

Enables AI assistants to interact with IBM watsonx.data lakehouses. Provides tools for querying data with SQL via Presto engines,...

zSkills Install

Author: Griffin Evans, Yemi Kelani, Zarin Ohiba, Anthony Randolph

IBM Z documentation and knowledge base MCP server. Provides real-time access to IBM documentation, GitHub repositories (IBM an...



Then install the second MCP server:

Settings | Settings | README.md

Search settings... Ctrl+F

- General
- Chat
- Editor
- Bob findings
- MCP
- Modes

Version: 1.0.0

Bob Docs

Report issue

Provides access to Redis key-value stores, enabling operations like setting, getting, deleting, and listing keys with optional...

A structured problem-solving tool that enables step-by-step analysis, thought revision, and branching logic for comple...

LSP-based code navigation server that provides precise symbol definitions, references, and structure analysis....

SonarQube Install
Author: SonarSource
SonarQube is an open-source platform for continuous inspection of code quality to perform automatic reviews with static...

SQLite Install
Author: modelcontextprotocol
Provides database interaction and business intelligence capabilities through SQLite, enabling SQL queries, data...

Terraform IBM Modules ... Install
Author: IBM Cloud
Provides structured access to the [Terraform IBM Modules (TIM)] (https://cloud.ibm.com/docs/ibm-cloud-...

Terraform Install
Author: hashicorp
The Terraform MCP Server provides seamless integration with Terraform Registry APIs, HCP Terraform, and...

Time Install
Author: modelcontextprotocol
Provides time and timezone conversion capabilities using IANA timezone names, with automatic system timezone...

Vault Install
Author: hashicorp
The Vault MCP Server provides integration with HashiCorp Vault for managing secrets and mounts, including...

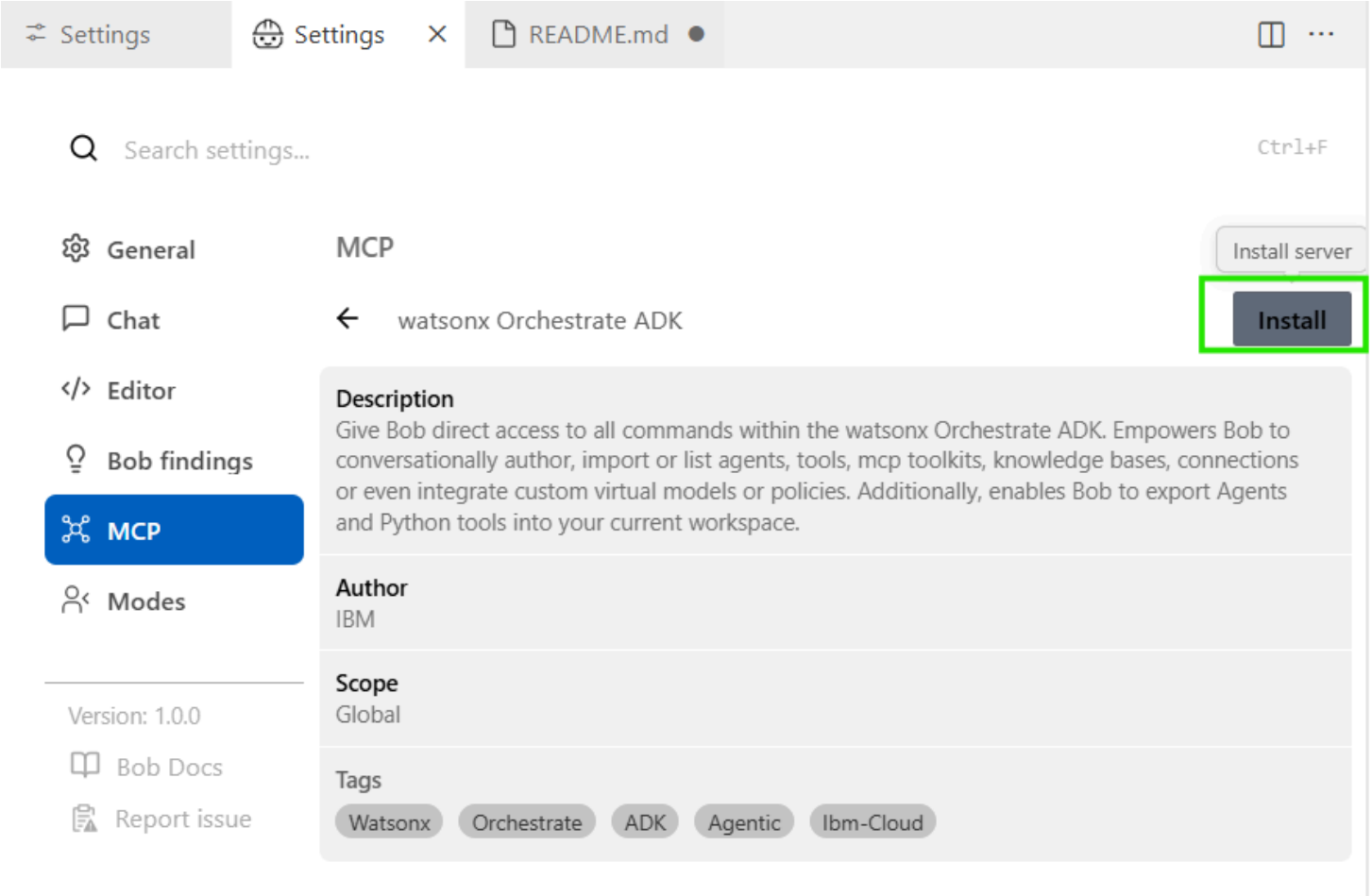
IBM Data Intelligence Install
Author: IBM
IBM Data Intelligence MCP Server provides modular and scalable integration with IBM Data Intelligence services,...

WatsonX Flows Install
Author: IBM
Enables integration with watsonx.ai Flows Engine, providing tools for Google Books and Wikipedia searches through a...

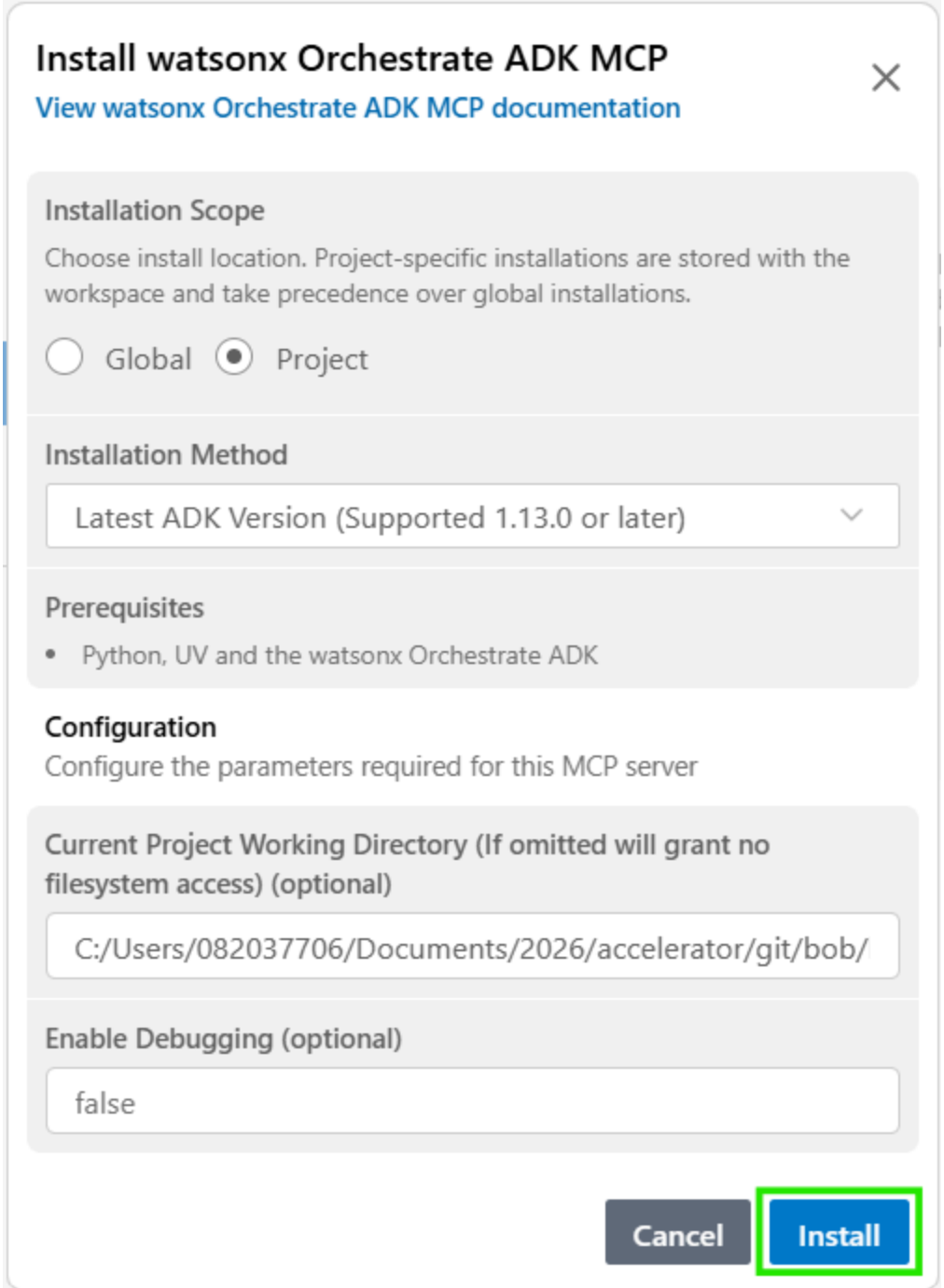
watsonx Orchestrate ADK Install
Author: IBM
Give Bob direct access to all commands within the watsonx Orchestrate ADK. Empowers Bob to conversationally auth...

IBM watsonx.data Install
Author: IBM
Enables AI assistants to interact with IBM watsonx.data lakehouses. Provides tools for querying data with SQL via Presto...

zSkills Install
Author: Griffin Evans, Yemi Kelani, Zarin Ohiba, Anthony Randolph
IBM Z documentation and knowledge base MCP server. Provides real-time access to IBM documentation, GitHub...



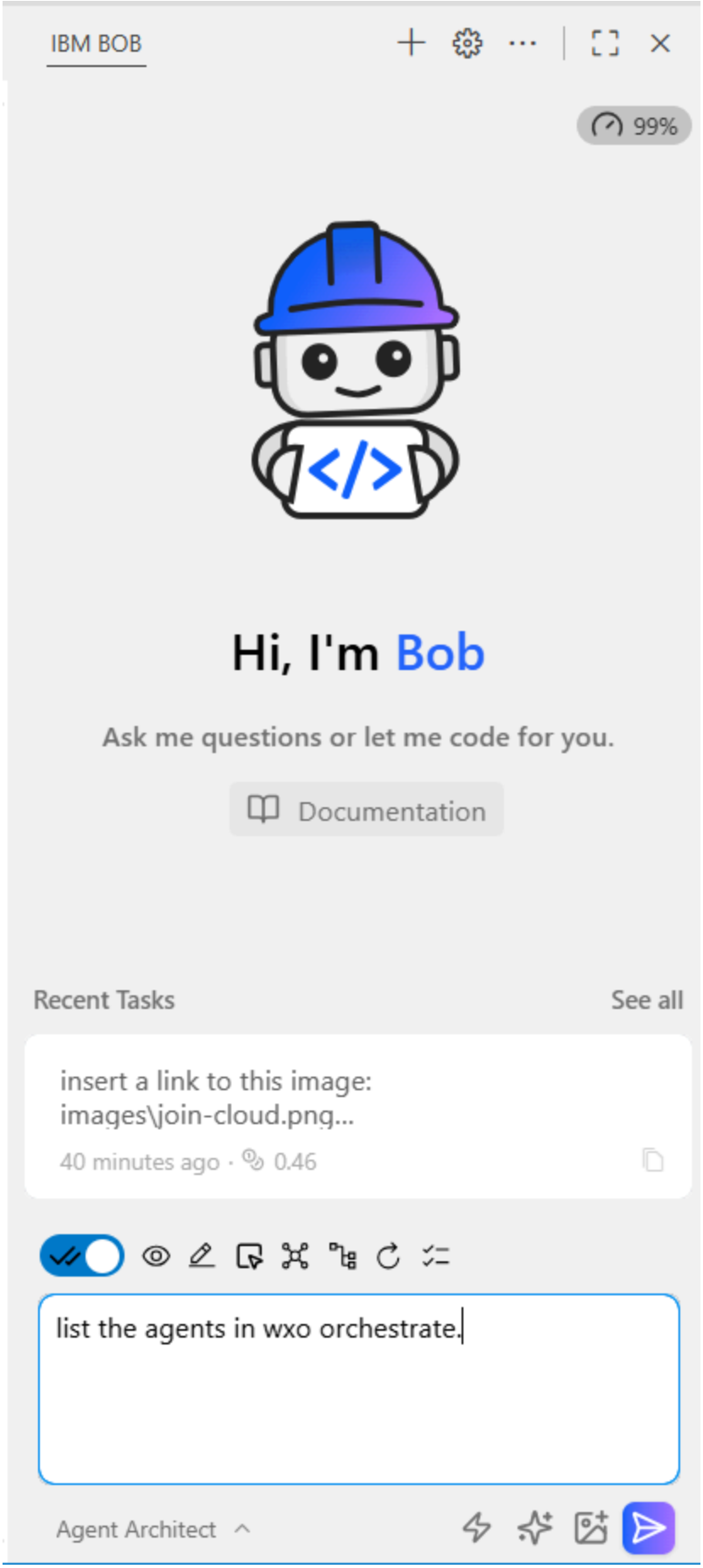
Then provide the path to your Bob repository where you and Bob will create the agent and tool files:



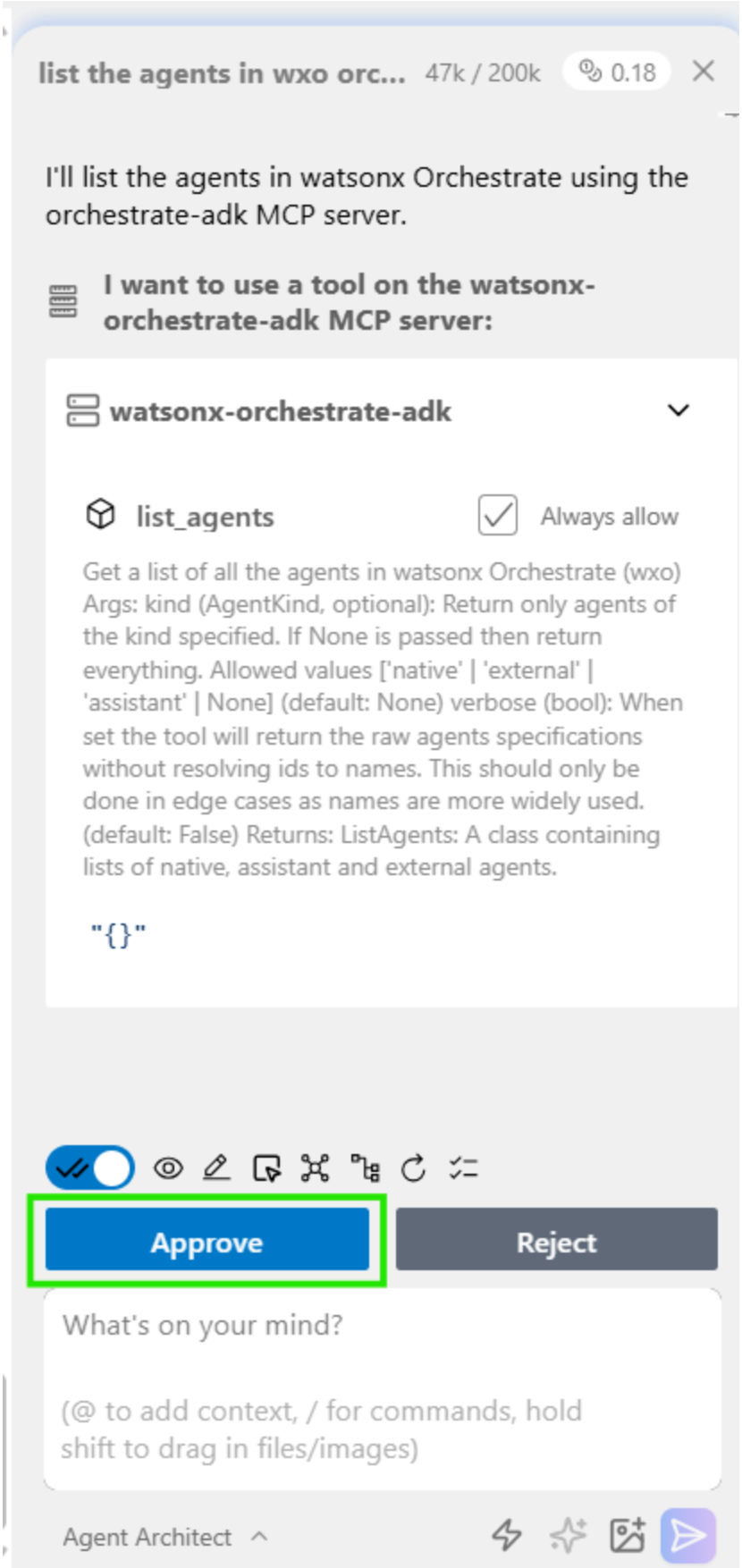
Remark:
For Windows, the correct format is something like this:
C:/Users/082037706/Documents/2026/accelerator/git/bob/bob-MCP-wxo/wxo-files

Test the ADK MCP server with a request like:

```
list the agents in orchestrate
```



Click "always allow" and "approve":



You will see the agents in wxo:

IBM BOB

list the agents in wxo orc... 48k / 200k 0.36

Native Agents:

- **AskOrchestrate**
  - ID: `2724102a-00d0-4384-813f-ec9082a93e83`
  - Description: A helpful AI assistant
  - LLM: `watsonx/meta-llama/llama-3-2-90b-vision-instruct`
  - Style: default
  - Status: Editable
  - Features:
    - Context access enabled
    - Knowledge base chat enabled (with document support)
    - Query rewrite enabled
    - No collaborators, tools, or toolkits currently configured

Assistant Agents: None

External Agents: None

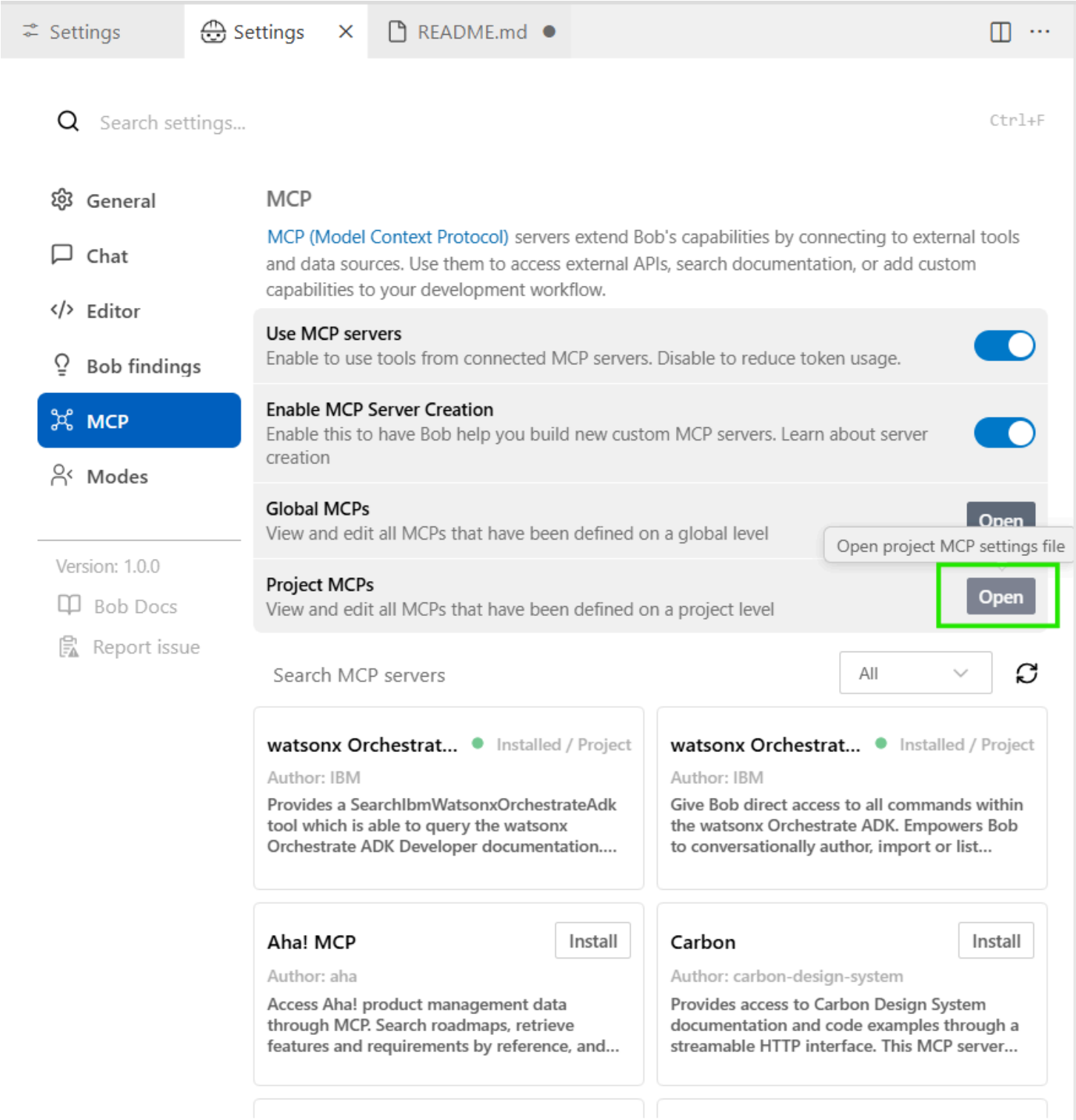
Start New Task

What's on your mind?

(@ to add context, / for commands, hold shift to drag in files/images)

Agent Architect

Install the 3rd wxo MCP server:



Add the following MCP server configuration in your mcp.json file.

(After the modification your file should look like : mcp.json)

```
"wxo-threads": {
    "command": "uvx",
    "args": [
        "--from",
        "wxo-threads-mcp-server",
        "--index-url",
        "https://test.pypi.org/simple/",
        "--extra-index-url",
        "https://pypi.org/simple/",
        "wxo-threads-mcp-server"
    ],
    "env": {
        "WXO_API_ENDPOINT": "https://api.us-south.watson-orchestrate.cloud.ibm.com/insta
        "WXO_API_KEY": "GTLI6TuscI1DBJd0EwQbJbTAzpNCrqAQA1JDKPj0ARSn",
        "WXO_DEFAULT_AGENT_ID": "2724102a-00d0-4384-813f-ec9082a93e83",
        "WXO_TIMEOUT": "30",
        "WXO_LOG_LEVEL": "DEBUG"
    },
    "timeout": 300,
    "disabled": false,
    "alwaysAllow": [
        "chat_with_agent",
        "get_last_agent_log"
    ]
}
```

You can test chatting with the default wxo agent (the agent ID provided in the MCP configuration file) with:
Ask the wxo agent: "3 + 3 = ?"

Create an Agent Using a Simple Tool Project with Bob in watsonx Orchestrate

Now that the 3 MCP servers are correctly configured, we will use them to create a small project with Bob.

Create a tool that performs a loan calculation (change the path wxo-files/tools/loan_tool.py according to your project - it has to be visible by the ADK MCP server):

```
Write a Python tool for IBM watsonx Orchestrate that exposes a single function:

def monthly_payment(principal: float, annual_rate: float, years: int) -> float:
    ...
Functional requirements:

The function calculates the fixed monthly payment for a loan (mortgage-style amortization).

Parameters:

principal: total loan amount (float).

annual_rate: annual interest rate in percent (e.g. 4.5 for 4.5%).

years: loan duration in years (integer).

Use the standard amortization formula with monthly compounding.

Correctly handle the edge case where annual_rate is 0 (no interest).

Return the monthly payment as a float rounded to 2 decimal places.

Raise a clear error (ValueError) for invalid inputs (negative or zero principal, non-positive ye

Code and structure requirements:

Use only the Python standard library (no external dependencies).

Add type hints and concise docstrings.

Organize the code as a watsonx Orchestrate tool:

Place the implementation in wxo-files/tools/loan_tool_XXXX.py where you have to replace XXXX wit

The file must define the monthly_payment function as the main callable entry point.
```

Now ask Bob to import the tool that has been created with something like:

```
import the python tool you have just created in wxo.
```

Then ask Bob to create an agent file that uses this tool (change the path if necessary):

```
create an agent file in wxo-files\agents directory.  this agent have to use the loan_tool tool y
```

Now import your newly created agent file in wxo with:

```
import the agent file you have just created in wxo.
```

Get the ID of your agent and use it to configure the 3rd MCP server:

```
What is the agent ID of the loan advisor agent you have just imported in wxo ?
```

Task Completed

The agent ID of the loan_advisor agent is:

`b3157eb9-f7e6-4061-b6ec-49a65f45c808`

Agent Details:

- **ID:** b3157eb9-f7e6-4061-b6ec-49a65f45c808
- **Name:** loan_advisor
- **Display Name:** Loan Payment Advisor
- **Status:** Active and editable
- **LLM:** watsonx/meta-llama/llama-3-2-90b-vision-instruct
- **Tools:** monthly_payment
- **Style:** default

The agent is successfully imported and ready to use in watsonx Orchestrate!

Start New Task

What's on your mind?

(@ to add context, / for commands, hold shift to drag in files/images)

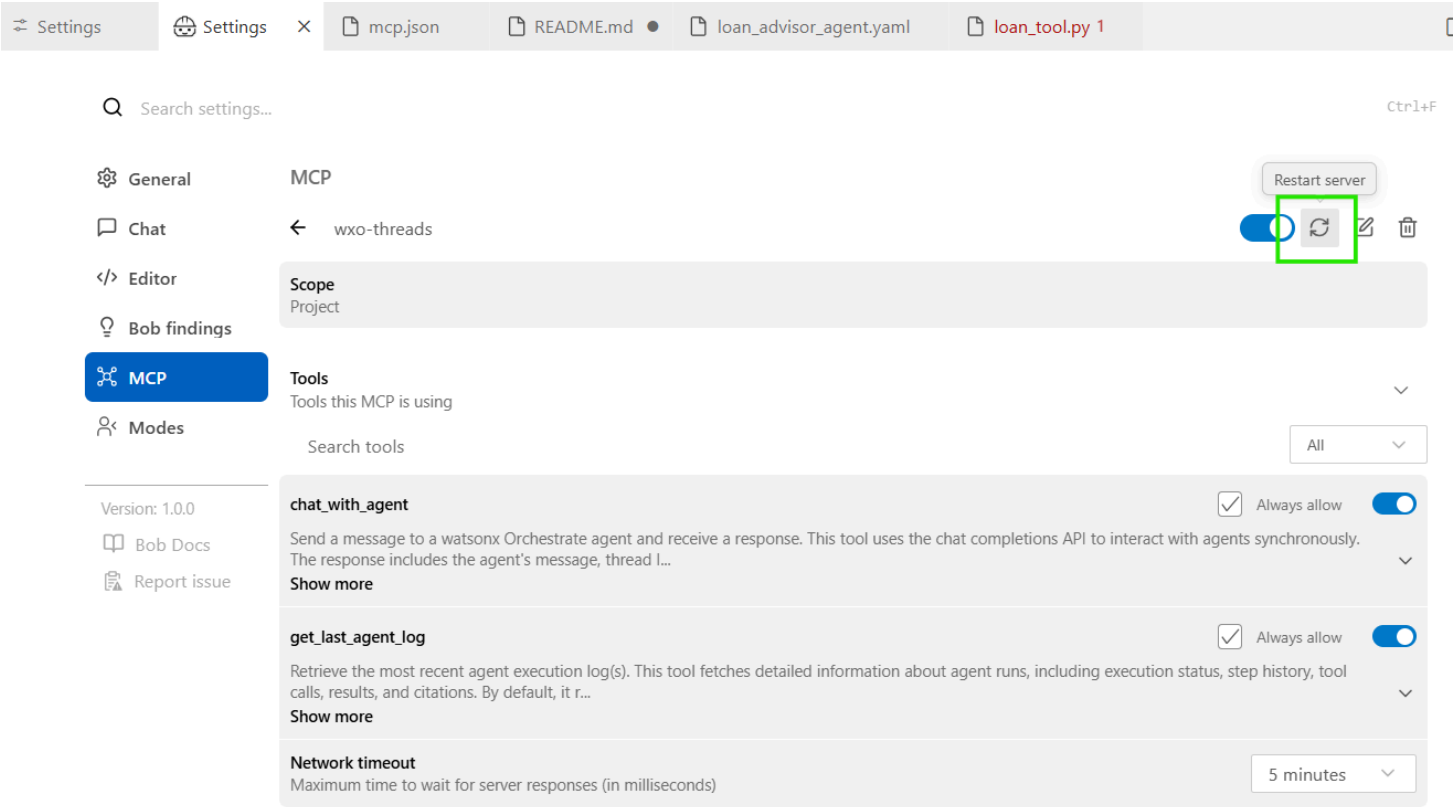
Agent Architect

Settings | Settings | mcp.json ● | README.md ● | loan_advisor_agent.yaml | loan_tool.py 1

.bob > mcp.json > {} mcpServers > {} wxo-threads > {} env > WXO_DEFAULT_AGENT_ID

```
2     "mcpServers": {
10        "watsonx-orchestrate-adk": {
19            "timeout": 300,
20            "alwaysAllow": [
21                "get_tool_template",
22                "import_tool"
23            ]
24        },
25        "wxo-threads": {
26            "command": "uvx",
27            "args": [
28                "--from",
29                "wxo-threads-mcp-server",
30                "--index-url",
31                "https://test.pypi.org/simple/",
32                "--extra-index-url",
33                "https://pypi.org/simple/",
34                "wxo-threads-mcp-server"
35            ],
36            "env": {
37                "WXO_API_ENDPOINT": "https://api.us-south.watson-orchestrate.cloud.ibm.com/instances/2dc5b062-6a29-4686-9
38                "WXO_API_KEY": "GTLI6IuscI1DBJd0EwQbJbIAzpNCrqAQA1JDKPj0ARSn",
39                "WXO_DEFAULT_AGENT_ID": "b3157eb9-f7e6-4061-b6ec-49a65f45c808",
40                "WXO_TIMEOUT": "30",
41                "WXO_LOG_LEVEL": "DEBUG"
42            },
43            "timeout": 300,
44            "disabled": false,
45            "alwaysAllow": [
46                "chat_with_agent",
47                "get_last_agent_log"
48            ]
49        }
50    }
51 }
```

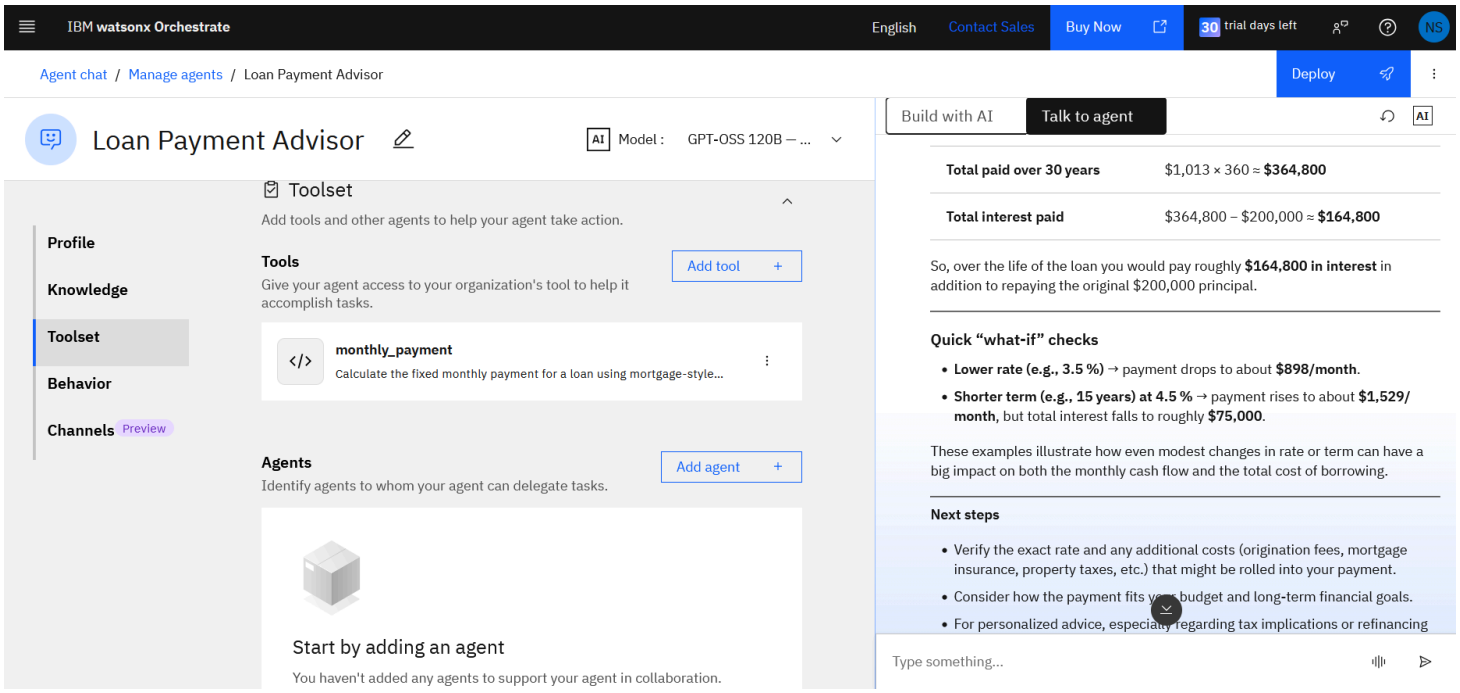
Restart the MCP server:



Now ask Bob to request the agent:

```
What would my monthly payment be for a $200,000 mortgage at 4.5% for 30 years?
```

And you can ask to get the logs from the last execution with the next command, it could be interesting to get these logs if there are errors during the agent request execution:

```
get the logs from the execution of the loan advisor agent.
```

You can also connect to the wxo UI to query the agent: